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MATHEMATICS TEST - CHAPTER 9: GEOMETRY AND POLYGONS

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**
Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Infinitely many lines can pass through	A single point	Two distinct points	Two parallel lines	Three points	●○○○
2. The branch proving geometry through reasoning is	Coordinate geometry	Practical geometry	Solid geometry	Demonstrative geometry	○○○●
3. Step of a proof that follows the statement is the	Proof	Corollary	Construction	Figure	○○○●
4. A result deduced directly from a theorem is a	Axiom	Conjecture	Corollary	Postulate	○○●○
5. Statements regarded true without proof are	Theorems	Corollaries	Conjectures	Fundamental assumptions	○○○●
6. In deductive reasoning we move from	Specific to general	Figure to proof	General to specific	Result to premise	○○●○
7. Which is a mathematical statement?	x + y = 9	x is less than 5	Sum of 2 and 5 is 7	Product of x and y is 10	○○●○
8. Through two different points pass	Two lines	Infinite lines	No line	Exactly one line	○○○●
9. A statement believed true but not proved is a	Theorem	Conjecture	Axiom	Corollary	○●○○
10. The logical chain from premises to conclusion is the	Argument	Premise	Corollary	Figure	●○○○
11. If a = b and b = c then a = c is the	Reflexive property	Symmetric property	Transitive property	Additive property	○○●○
12. A number is always equal to itself. This is the	Reflexive property	Transitive property	Symmetric property	Closure property	●○○○
13. The word "axiom" comes from the Greek word	Theorien	Geometria	Postulatum	Axioma	○○○●
14. In the "Given" step, information is written	As a conclusion	As a corollary	Symbolically	Without symbols	○○●○

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Questions	A	B	C	D	A B C D
15. An assumption that something is true is the	Conclusion	Argument	Proof	Premise	○○○●
16. The word "theorem" comes from the Greek word	Theorien	Axioma	Geometria	Polygonos	●○○○
17. Interchanging given and to-prove gives the	Corollary	Converse	Axiom	Conjecture	○●○○
18. A proof is a series of conjectures, axioms and	Open sentences	Unproved guesses	Proved theorems	Diagrams only	○○●○
19. Using a conjecture, the next term in 10, 20, 30, 40, ... is	50	49	53	52	●○○○
20. Using a conjecture, the next term in 1, 2, 4, 8, 16, ... is	30	32	36	27	○●○○
21. For x = 10, the value of $(x + 1)^2 + 5$ is	125	126	124	132	○●○○
22. The mathematical statement "Square root of 36 is 6" is	True	False	Open sentence	Not a statement	●○○○
23. The mathematical statement "Sum of 2 and 5 is 7" is	False	Open sentence	Not a statement	True	○○○●
24. Using a conjecture, the next term in 3, 7, 11, 15, ... is	19	20	21	22	●○○○
25. For x = 10, the value of $x^2 + 2x + 6$ is	126	132	128	125	●○○○
26. Value of $(103)^2$ is	10600	10709	10306	10609	○○○●
27. Using a conjecture, the next term in 7, 14, 21, 28, ... is	35	33	39	34	●○○○
28. The mathematical statement "Square of 12 is 144" is	False	Open sentence	True	Not a statement	○○●○
29. If ∠A = ∠D and ∠B = ∠E, then ΔABC ~ ΔDEF by	SSS congruence	AA similarity	RHS	SAS congruence	○●○○
30. If DE ∥ BC in ΔABC, then AD/DB equals	AE/EC	AE/AC	DE/BC	AB/AC	●○○○
31. Corresponding sides of similar figures are	Equal	Parallel	Proportional	Perpendicular	○○●○

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Questions	A	B	C	D	A B C D
32. All squares are	Congruent to one another	Never similar	Similar to one another	Not rectangles	○○●○
33. A square and a non-square rectangle are	Similar	Not similar	Congruent	Equal	○●○○
34. Two equiangular triangles are	Similar	Congruent	Equal in area	Right angled	●○○○
35. Similar polygons have equal corresponding angles and	Equal sides	Equal areas	Proportional sides	Equal diagonals	○○●○
36. Two equilateral triangles are always	Congruent	Right angled	Similar	Scalene	○○●○
37. ΔABC ~ ΔDEF, AB = 5 cm, DE = 7.5 cm, BC = 12 cm. Then EF =	20 cm	18 cm	8 cm	22 cm	○●○○
38. ΔABC and ΔDEF have AB/DE = BC/EF = CA/FD = 2, AB = 6, BC = 8, CA = 4. DE =	12 cm	3 cm	6 cm	4 cm	○●○○
39. AD = 3, DB = 5, AE = 6, EC = 10. Is DE ∥ BC?	No	Cannot say	Only if AB = AC	Yes	○○○●
40. AD = 5, DB = 7, AE = 4, EC = 8. Is DE ∥ BC?	Yes	Cannot say	No	Only if AB = AC	○○●○
41. ΔABC ~ ΔDEF, AB = 3 cm, DE = 4.5 cm, BC = 4 cm. Then EF =	2.67 cm	8 cm	6 cm	10 cm	○○●○
42. AD = 6, DB = 9, AE = 4, EC = 6. Is DE ∥ BC?	No	Yes	Cannot say	Only if AB = AC	○●○○
43. AD = 4, DB = 6, AE = 2, EC = 3. Is DE ∥ BC?	No	Yes	Cannot say	Only if AB = AC	○●○○
44. DE ∥ BC; AD = 4, DB = 5, AE = 8. Then EC =	10	11	9	12	●○○○
45. ΔABC and ΔDEF have AB/DE = BC/EF = CA/FD = 2, AB = 6, BC = 8, CA = 4. FD =	3 cm	8 cm	2 cm	1 cm	○○●○
46. ΔXYZ is equilateral and LM ∥ YZ. Then ΔXLM is	Scalene	Equilateral	Right angled	Isosceles obtuse	○●○○
47. AD = 3, DB = 5, AE = 6, EC = 9. Is DE ∥ BC?	Yes	Cannot say	No	Only if AB = AC	○○●○
48. DE ∥ BC; AD = 2, DB = 5, AE = 6. Then EC =	14	15	30	17	○●○○

Questions	A	B	C	D	A B C D
49. Two cuboids are similar if corresponding edges are	Equal	Parallel	Proportional	Perpendicular	○○●○
50. If the scale factor is k, volume ratio is	k^2	k^3	k	3k	○●○○
51. Triangles with common height have area ratio equal to ratio of their	Squares of bases	Bases	Perimeters	Angles	○●○○
52. Same-density similar solids: mass ratio equals	Cube of length ratio	Square of length ratio	Length ratio	Area ratio	●○○○
53. If the scale factor is k, area ratio is	k^2	k	k^3	2k	●○○○
54. Base area of smaller cylinder is 240 cm^2; ratio 9:4. Larger base area is	320 cm^2	135 cm^2	480 cm^2	540 cm^2	○○○●
55. Sides of two similar figures are 3:4. Ratio of their areas is	3:16	27:64	9:4	9:16	○○○●
56. Areas of bases of similar cylinders are 9:4. Ratio of their heights is	3:2	9:4	81:16	2:3	●○○○
57. Solids A, B are similar with scale factor 2:3. Ratio of volumes is	27:8	8:27	2:3	4:9	○●○○
58. Triangles with bases 3 cm and 7 cm and common height have area ratio	3:10	9:49	7:3	3:7	○○○●
59. Areas of similar solids are 16:25. Ratio of their volumes is	4096:15625	256:625	16:25	64:125	○○○●
60. Areas of similar solids are 9:49. Ratio of their volumes is	27:343	3:7	81:2401	9:49	●○○○
61. Similar solids have lengths 3:5. First volume is 27 cm^3. Second volume is	75 cm^3	133 cm^3	120 cm^3	125 cm^3	○○○●
62. Sides of two similar figures are 4:7. Ratio of their areas is	4:7	16:49	4:49	16:7	○●○○
63. Spheres have radii 6 cm and 8 cm. Ratio of their areas is	6:8 only	27:64	9:16	3:4	○○●○
64. Spheres have radii 6 and 8. Larger area is 1600 cm^2. Smaller area is	1000 cm^2	600 cm^2	800 cm^2	900 cm^2	○○○●
65. Areas of similar solids are 25:36. Ratio of their volumes is	125:216	25:36	15625:46656	625:1296	●○○○

Questions	A	B	C	D	A B C D
66. Lengths of two similar solids are in ratio 4:5. Ratio of volumes is	4:125	64:25	64:125	16:25	○○●○
67. Areas of similar solids are 1:4. Ratio of their volumes is	1:64	1:2	1:16	1:8	○○○●
68. Solids A, B are similar with scale factor 2:3. Ratio of surface areas is	8:27	16:81	4:9	2:3	○○●○
69. A polygon is a two-dimensional	Curved figure	Open figure	Closed figure of straight sides	Solid figure	○○●○
70. Sum of interior angles of a quadrilateral is	180°	540°	720°	360°	○○○●
71. The point where two sides of a polygon meet is a	Apothem	Diagonal	Centre	Vertex	○○○●
72. Area of a regular polygon with perimeter P, apothem a is	P × a	P + a	½ × P × a	½ × P/a	○○●○
73. An exterior angle is formed by a side and the extension of its	Opposite side	Diagonal	Adjacent side	Apothem	○○●○
74. A regular polygon is also called	Scalene	Equiangular and equilateral	Concave	Irregular	○●○○
75. The regular polygon with fewest sides is	Square	Pentagon	Rhombus	Equilateral triangle	○○○●
76. The apothem is also the radius of the	Inscribed circle	Circumscribed circle	Sector	Segment	●○○○
77. Triangles formed from one corner of an n-gon number	n − 2	n − 1	n − 3	n	●○○○
78. A polygon with one unequal side or angle is	Regular	Irregular	Equiangular	Equilateral	○●○○
79. Interior angles of a polygon lie	Outside the polygon	On the diagonals	At the centre	Inside the polygon	○○○●
80. Sum of exterior angles of any polygon is	180°	360°	540°	720°	○●○○
81. The triangle is widely used in construction because it is	Weak	Costly	Irregular	Strong	○○○●
82. The polygon with equal interior and exterior angles is	Square	Triangle	Pentagon	Hexagon	●○○○

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Questions	A	B	C	D	A B C D
83. Number of diagonals of an n-sided polygon is	n(n − 3)/2	n(n − 1)/2	n(n − 2)/2	n(n + 3)/2	●○○○
84. Area of a triangle is	Base × height	½ × base × height	Base + height	2 × base × height	○●○○
85. Sum of interior angles of an n-sided polygon is	(n − 1) × 180°	(n − 2) × 180°	n × 180°	(n + 2) × 180°	○●○○
86. Opposite sides of a parallelogram are	Equal only	Perpendicular	Equal and parallel	Unequal	○○●○
87. In the word polygon, "gon" means	Angles	Sides	Lines	Points	●○○○
88. Triangles formed by joining diagonals from one vertex of a 9-gon:	8	7	9	6	○●○○
89. Interior angle of a polygon is 5 times its exterior angle. Number of sides is	15	10	8	12	○○○●
90. Each exterior angle of a regular polygon is 10°. Number of sides is	37	36	38	72	○●○○
91. Sum of interior angles of a 9-sided polygon is	1620°	900°	1440°	1260°	○○○●
92. Each interior angle of a regular 36-gon is	170°	160°	20°	10°	●○○○
93. Each interior angle of a regular 18-gon is	10°	20°	160°	166°	○○●○
94. Sum of interior angles of a polygon with 20 sides is	3960°	6480°	3240°	3060°	○○●○
95. Smallest interior angle of a regular polygon is	120°	30°	90°	60°	○○○●
96. A convex polygon has 14 diagonals. Number of sides is	14	6	8	7	○○○●
97. Each exterior angle of a regular 36-gon is	20°	15°	10°	5°	○○●○
98. Each exterior angle of a regular triangle is	60°	120°	130°	59°	○●○○
99. A hexagonal plot has side 10 m. Perimeter is	36 m	100 m	50 m	60 m	○○○●

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Questions	A	B	C	D	A B C D
100. Sum of interior angles of a 17-sided polygon is	2340°	2520°	2700°	2880°	○○●○
101. A polygon with 7 sides is called a	heptagon	nonagon	decagon	pentagon	●○○○
102. Sum of interior angles of a polygon with 16 sides is	2700°	5040°	2520°	360°	○○●○
103. Each interior angle of a regular polygon is 140°. Number of sides is	18	7	13	9	○○○●
104. Perimeter of a regular polygon with 10 sides, each 5 cm, is	15 cm	50 cm	45 cm	25 cm	○●○○
105. Each interior angle of a regular 24-gon is	177°	165°	171°	30°	○●○○
106. Each exterior angle of a regular 30-gon is	6°	17°	24°	12°	○○○●
107. Each interior angle of a regular polygon is 162°. Number of sides is	21	40	20	19	○○●○
108. Sum of interior angles of a polygon with 10 sides is	1800°	360°	1260°	1440°	○○○●
109. A regular polygon has perimeter 42 cm and apothem 5 cm. Its area is	210 cm^2	212 cm^2	47 cm^2	105 cm^2	○○○●
110. Each interior angle of a regular polygon is 108°. Number of sides is	4	10	5	6	○○●○
111. A polygon with 6 sides is called a	octagon	pentagon	heptagon	hexagon	○○○●
112. Sum of all interior angles is 9 right angles. Such a polygon	Is a hexagon	Is a heptagon	Is a pentagon	Does not exist	○○○●
113. A convex polygon has 54 diagonals. Number of sides is	11	12	42	10	○●○○
114. Sum of interior angles of a polygon is 360°. Number of sides is	8	4	2	6	○●○○
115. Interior angle of a regular polygon with 16 sides is	157.5°	168°	150°	135°	●○○○
116. A convex polygon has 27 diagonals. Number of sides is	10	7	18	9	○○○●

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Questions	A	B	C	D	A B C D
117. A polygon with 4 sides has how many diagonals?	2	6	4	14	●○○○
118. Triangles formed by joining diagonals from one vertex of a 12-gon:	12	10	13	9	○●○○
119. A polygon with 10 sides is called a	hexagon	decagon	nonagon	octagon	○●○○
120. Triangles formed by joining diagonals from one vertex of a 13-gon:	13	12	11	14	○○●○
121. A regular polygon has perimeter 20 cm and apothem 3 cm. Its area is	40 cm^2	30 cm^2	62 cm^2	60 cm^2	○●○○
122. Each exterior angle of a regular polygon is 18°. Number of sides is	22	20	18	19	○●○○
123. Wall around a lawn of perimeter 15 m costs Rs 220/m. Cost is	Rs 3300	Rs 3500	Rs 2200	Rs 3000	●○○○
124. A regular polygon has perimeter 36 cm and apothem 5 cm. Its area is	90 cm^2	45 cm^2	182 cm^2	41 cm^2	●○○○
125. Four angles of a pentagon are 70°, 85°, 135°, 90°. The remaining angle is	155°	160°	150°	170°	○●○○
126. Exterior angle of a regular polygon is 120°. Polygon is	Square	Pentagon	Equilateral triangle	Hexagon	○○●○
127. Five angles of a hexagon are 135°, 125°, 150°, 70°, 145°. The remaining angle is	85°	100°	105°	95°	○○○●
128. A convex polygon has 90 diagonals. Number of sides is	16	17	75	15	○○○●
129. Five angles of a hexagon are 120°, 110°, 150°, 105°, 90°. The remaining angle is	165°	145°	155°	150°	○●○○
130. Diagonals from one vertex of a polygon with 14 sides:	15	12	11	13	○○●○
131. Exterior angles of a quadrilateral are in ratio 2:3:5:8. The largest is	184°	160°	20°	148°	○●○○
132. Diagonals from one vertex of a polygon with 10 sides:	7	9	8	10	●○○○
133. A polygon with 9 sides is called a	nonagon	pentagon	heptagon	octagon	●○○○

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Questions	A	B	C	D	A B C D
134. Each interior angle of a regular dodecagon is	162°	140°	150°	156°	○○●○
135. Perimeter of a regular polygon with 6 sides, each 5 cm, is	40 cm	11 cm	30 cm	36 cm	○○●○
136. Points equidistant from three non-collinear points number	One	Two	Three	Infinite	●○○○
137. Locus of points at fixed distance from a point (plane) is a	Line	Parabola	Square	Circle	○○○●
138. Two lines meeting at 60°: bisector makes with each	30°	60°	15°	45°	●○○○
139. Locus of points at distance a from line l is	Two parallel lines	One line	A circle	A perpendicular line	●○○○
140. Every point on the locus	Is on the x-axis	Is outside the figure	Satisfies the condition	Is collinear	○○●○
141. Common base AB of isosceles triangles XAB, YAB: XY is	Perpendicular bisector of AB	Parallel to AB	Equal to AB	Median only	●○○○
142. Points 5 m from line l can lie on how many paths?	2	1	3	4	●○○○
143. Point equidistant from three non-collinear points is	Incentre	Circumcentre	Centroid	Orthocentre	○●○○
144. Method of intersecting loci finds points that	Satisfy several conditions	Lie on a line	Are collinear	Form a circle only	●○○○
145. Locus of points 4 units from the x-axis is	$y = 4$ only	$y = 4$ and $y = -4$	$x = 4$ only	$x = \pm 4$	○●○○
146. Points A, B on axes at 4.5 cm each; locus equidistant from origin is a	Circular arc (radius 4.5 cm)	Straight line	Parabola	Square	●○○○
147. Points 6 cm from line l form two parallel lines. Distance between them is	13 cm	6 cm	18 cm	12 cm	○○○●
148. Locus of points 10 cm from a fixed point O in a plane is a circle of radius	10 cm	14 cm	20 cm	5 cm	●○○○
149. Two parallel lines are 4 cm apart. A line midway is at distance from each	3 cm	2 cm	4 cm	1 cm	○●○○
150. AB = 6 cm. The perpendicular bisector of AB meets it at a distance from A of	2 cm	4 cm	3 cm	1.5 cm	○○●○

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